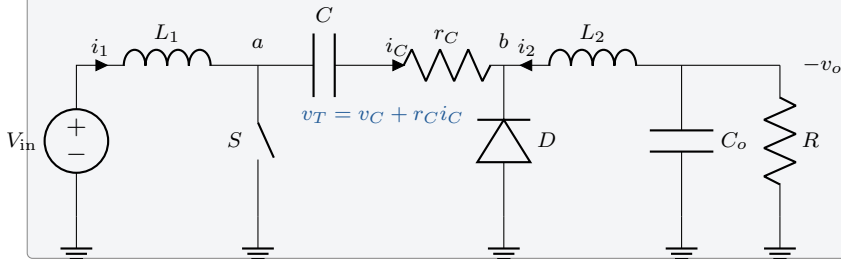
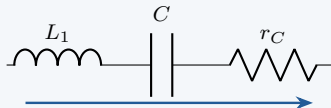


(a) Classical inverting Ćuk power stage and adopted signs

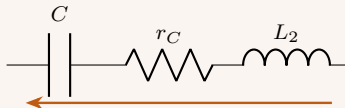


(b) $u = 0$: $i_C = +i_1$



switch open; diode active
positive port-current direction

(c) $u = 1$: $i_C = -i_2$



switch path active; diode blocked
capacitor-port current reverses

v_C is continuous at the transition; the signed ESR drop changes.